Groundwater source assessment program for the state of Hawaii, USA: methodology and example application

Robert B. Whittier · Kolja Rotzoll · Sushant Dhal · Aly I. El-Kadi · Chittaranjan Ray · Daniel Chang

Abstract The 1996 reauthorization of the Safe Drinking Water Act required that each state in the US addresses the protection of public drinking water sources, including the development and implementation of a source-water assessment program. Such a program includes delineating source-water assessment areas, inventorying potential contaminant sources within this area, and determining the water system's susceptibility to contamination. The public was also involved in various phases of the program. Hawaii's groundwater source assessment program is presented, along with an approach for implementation, which is consistent with federal requirements. The approach integrates groundwater models, aquifer databases, and a geographic information system. Source assessment areas were delineated by using numerical groundwater-flow models that used site-specific data to their fullest availability. The proposed approach is flexible enough to allow easy future updates as more sources are identified or as new information becomes available. The final product includes numerical scores that quantify the relative source susceptibility to contamination. Aquifer models developed in this study are potentially useful for future site-specific protection efforts or for other modeling purposes.

Keywords Numerical modeling · Groundwater management · Water-resources conservation · Groundwater protection · USA

Received: 20 April 2009 / Accepted: 19 October 2009
Published online: 25 November 2009

© Springer-Verlag 2009

R. B. Whittier · K. Rotzoll · A. I. El-Kadi (✉)
Department of Geology & Geophysics and Water Resources Research Center,
University of Hawaii,
2540 Dole St., Holmes Hall 283, Honolulu, HI 96822, USA
e-mail: elkadi@hawaii.edu
Tel.: +1-808-9566331
Fax: +1-808-9565512

S. Dhal · C. Ray
Department of Civil Engineering,
University of Hawaii,
1680 East-West Road, Honolulu, HI 96822, USA

D. Chang
State of Hawaii Department of Health,
Safe Drinking Water Branch,
119 Ala Mona Blvd, Honolulu, HI 96814, USA

Introduction

The 1996 reauthorization of the US Federal Safe Drinking Water Act required that each state in the US of America focuses on the protection of drinking-water sources through technical assistance to public water systems, sound assessment methods, and public participation and access to information. The reauthorization act also provided funding and listed specific requirements and deadlines for the development and implementation of source water assessment programs (SWAP). Although source water protection was not required by the reauthorization act, states were encouraged to establish protection measures and link SWAP with source water protection.

Source water assessment of a public water system is aimed at evaluating its vulnerability to contamination and is completed through four steps: (1) delineating the source water-assessment area, (2) preparing an inventory of potential contaminant sources within such an area, (3) determining the water system's susceptibility to contamination, and (4) insuring public participation and outreach.

The US Environmental Protection Agency (EPA) identified five criteria that could be used for the purposes of wellhead area delineation (US EPA 2006): (1) fixed distance away from the source, (2) area subjected to drawdown due to pumping, (3) time of travel (TOT) of contaminants reaching the source, (4) flow boundaries, and (5) soil and water assimilative capacity. Of these five criteria, Oki et al. (1992) concluded that three (fixed distance, TOT, and flow boundaries) are potentially suitable for use in Hawaii. Uncertainties in addressing the issue of assimilative capacity and the need for site specific and chemical data reduce usefulness of this concept. The drawdown criterion is not useful for cases

 DOI 10.1007/s10040-009-0548-6

where the area of water contribution to a well differs from the area affected by drawdown. In addition, high conductivity of Hawaii aquifers causes relatively small drawdowns with expected significant measuring errors.

States have adopted approaches of varied degrees of complexity for developing SWAP. For example, Virginia's Department of Health determined that a fixed radius delineation approach is suitable for the state, which has more than 3,000 sources (Virginia Department of Health 2007). In addition to technical and logistical considerations, and as is the case for most states, justifications included the fact that neither water purveyors nor the Department of Health have adequate staff and financial resources to complete more complex delineations in the allotted time frame.

The California SWAP, serving more than 15,000 sources, recommends four mandatory delineated areas for groundwater sources, and an optional fifth area (California Department of Public Health 2007). Both fixed radius and travel time methods were used in the assessment. Based on simplified approaches, charts are provided for estimating the sizes of zones based on the travel time method. Additional buffer zone can be used to delineate a larger setback away from activities that may be significant potential sources of contamination (e.g., landfills or hazardous material disposal sites), and to provide additional information that may be helpful for longer-term planning.

The New Jersey SWAP, which includes more than 6,000 sources, recommends using a combination of a groundwater model and the fixed radius method (New Jersey Department of Environmental Protection 2005). Uncertainty in head gradient and groundwater flow direction is accounted for by expanding the capture zone by adding two additional zones of equal sizes, one on each side. These are estimated by rotating the originally estimated zone 20° clockwise and another 20° counterclockwise with the well as a pivot point. If appropriate data exists, the New Jersey SWAP recommends the use of physically based two- or three-dimensional models.

The objective of this study was to develop and implement a groundwater source assessment program for the state of Hawaii. The approach differs from those used by other states by applying detailed groundwater models for all sources. The models employed site-specific hydrogeological data to the best of their availability and can allow future updates as more information becomes available. The linkage between the models and geographic information system (GIS) data allowed utilization of spatially distributed data and presentation of assessment maps and susceptibility results in a convenient manner. The developed aquifer models can serve as an excellent start for future source protection studies that will deal with site-specific water systems. In addition, the study emphasizes areas of needed research and provides a framework towards a better understanding of Hawaii groundwater resources. Following the description of the approach adopted for SWAP, an example application is presented for the Island of Oahu.

Methods

Groundwater sources provide about 99% of Hawaii's public water use and 50% of all freshwater used in the state (Gingerich and Oki 2000). At the time of this study, the groundwater comprised 405 sources out of 453 total sources. The rest is served by surface-water sources and groundwater sources under the direct influence of surface water such as spring sources and water-development tunnels. This report only addresses approaches adopted for groundwater sources. Approaches and applications for those related to surface water are documented in Whittier et al. (2006a). The assessment program for groundwater sources developed in this study is consistent with federal mandates and requirements, and with the objectives of existing assessment and protection efforts such as the Hawaii Wellhead Protection Program (WHPP; US EPA 2006). Public participation was achieved through meetings with concerned communities throughout program development and the final review process.

Delineating assessment zones

The WHPP defined two general zones that require delineation: (1) a near source zone that provides protection against direct introduction of contaminants through and around the well casing (via vandalism, chemical spills, wastewater overflows, etc.), and (2) a larger source zone that is based either on simulated capture zones or that is limited by aquifer physical boundaries. This zone reflects concerns of past or potential incidences of contamination resulting from chemical spills or natural outflows from existing contamination sources. Both, fixed distance and TOT criteria, were used for delineating individual zones within the assessment area in the SWAP approach. The following zones were delineated for groundwater sources.

- Zone A: A circular area around the well of a fixed 17-m radius.
- Zone B: An area that contributes groundwater to the well over a 2-year period. The 2-year TOT is the current recommendation for the proposed Groundwater Rule (US EPA 2006) and is based on conservative survival times for bacteria and viruses in soil and groundwater.
- Zone C: An area that contributes groundwater to the well over a 10-year time period for all islands. The 10-year TOT threshold was chosen to allow sufficient time to implement management and remedial measures to mitigate contamination from accidental contaminant spills or other causes. For Molokai, an additional 25-year TOT was chosen in accordance with the Molokai WHPP Demonstration Project (Horsely and Witten Inc. 1992). Participants in a SWAP public information meeting held on Molokai on 24 September 1998 indicated a preference for a more conservative approach by retaining the use of the 25-year capture-zone delineation.

 DOI 10.1007/s10040-009-0548-6

Hydrogeology

The numerical models used in this study to delineate zones B and C required detailed information about hydrogeological conditions in the islands including recharge, groundwater withdrawal, and aquifer parameters. Disparity in aquifer types and dominance of hydrogeological variability complicated modeling effort. Extensive coverage of Hawaii geology and hydrogeology can be found in, e.g., Macdonald et al. (1983), Hunt (1996), and Lau and Mink (2006). Ranges of hydraulic conductivities and other parameters provided in these studies were used as guidelines in the model calibration process.

The Islands of Hawaii are characterized by high rainfall and highly permeable aquifers. Groundwater generally occurs in a freshwater-lens or high-level systems. High-level groundwater is impounded by vertical flow barriers such as volcanic dikes, which exist abundantly in the rift zones of the volcanoes. Groundwater also exists on Oahu as high-level water beneath the Schofield Plateau between hydrologic barriers of uncertain origin (Fig. 1) and locally as perched water over poorly permeable layers. The groundwater elevations can reach several tens to hundreds of meters above sea level in the high-level systems (Lau and Mink 2006). Groundwater in the freshwater-lens aquifers occurs in highly conductive flank lava flows, which are typically several tens of meters thick, embedded with coarser clinker beds. The flank lavas are overlain by coastal sediments of variable thickness up 5 km in Pearl Harbor area. The sedimentary wedge is usually referred to as caprock because it restricts discharge to the ocean and acts as a semi-confining layer. In the transversal flow direction, alluvial valley-fill deposits impede lateral groundwater movement between stream valleys (Oki 2005). The hydraulic head of the freshwater-lens groundwater rarely exceeds a few meters above sea level and the hydraulic gradient is about 0.2 m/km, although local variations in the hydraulic gradient may be caused by aquifer heterogeneities, discharging springs, heavily pumped wells, valley-fill or other barriers.

Modeling procedure

Zones B and C were delineated using the numerical groundwater models MODFLOW 2000 (Harbaugh et al. 2000) and MODPATH (Pollock 1994). MODFLOW is a three-dimensional, cell-centered, finite difference, saturated groundwater-flow model developed by the US Geological Survey (USGS). MODPATH was used to simulate the conservative advection of dissolved phase contaminants or microbes within the groundwater system over the selected periods ignoring the effect of dispersion.

Needed for the numerical solutions are values of hydrogeologic parameters and values of heads or flows along the boundaries of the model. MODPATH uses hydraulic heads from MODFLOW and computes pore-water velocity using the hydraulic conductivity and porosity data. The use of the three-dimensional models easily allowed the consideration of layer heterogeneity and partial penetration of the well screens. MODFLOW and MODPATH have also been incorporated in the package Groundwater Modeling System (EMS-i 2006),

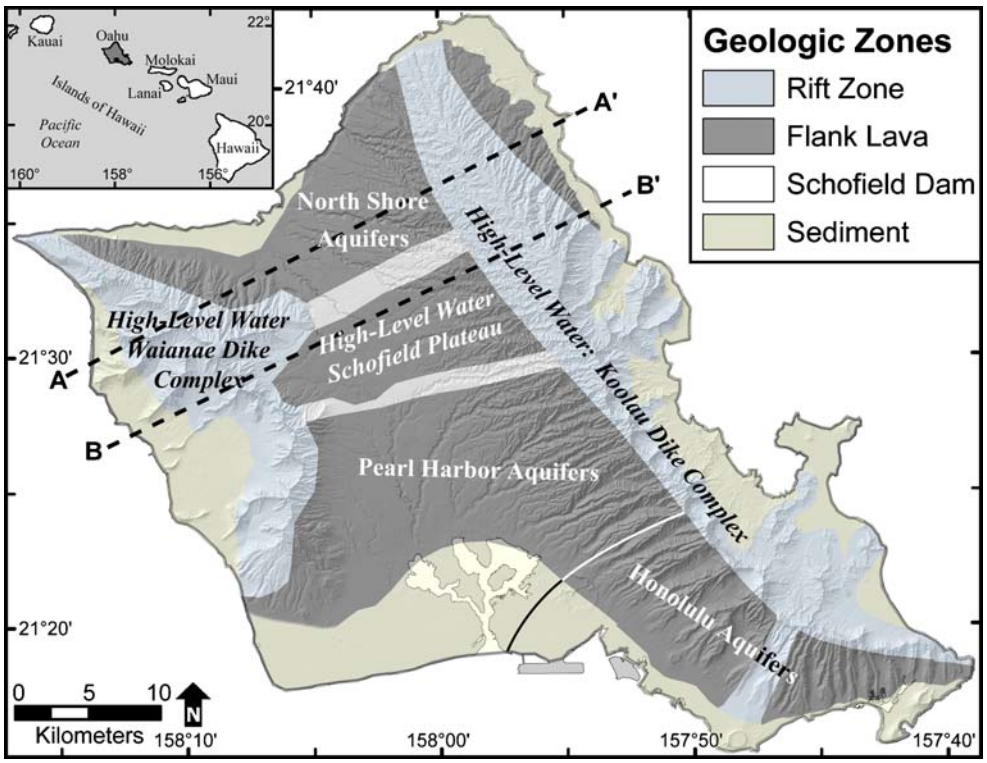


Fig. 1 Hydrogeologic zones used in the numerical model characterizing the Island of Oahu, Hawaii and traces of the cross sections A–A' and B–B' discussed in the following section

 DOI 10.1007/s10040-009-0548-6

easing the incorporation of electronic data such as GIS shape and raster files.

Two approaches were tested in modeling the dike complexes and valley-fill zones, which are characterized by a heterogeneous nature and low conductivity values. First, the dike complexes were modeled as an equivalent porous medium with a very low average hydraulic conductivity to obtain a reasonable match between measured and estimated groundwater elevations. Dikes occurring in well-defined rift zones commonly have a preferred orientation that parallels the axis of the rift zone. In these areas, a second approach was taken that utilized horizontal flow barriers (HFB). HFBs simulate physical barriers such as slurry trenches. They are modeled as vertical arcs that extend downward the entire thickness of the respective layer. HFBs are assigned a hydraulic-conductance value, which is the ratio of hydraulic conductivity divided by the width of the barrier. HFBs were also used to simulate low-permeability valley-fill deposits occurring in narrow steep-walled valleys.

Streams cut many rift zones and dike complexes. Due to the elevated groundwater elevation in these areas, many streams intersect the water table resulting in large discharge of groundwater to surface water. The streams were thus modeled as drains. The amount of water removed by the drain is dependent on the head difference between the groundwater and the drain elevation, the conductance of the drain, and the length of the draining arc. The conductance value, which is estimated through calibration, is a lumped parameter representing the hydraulic conductivity and the thickness of the streambed materials. Since only the elevation of streams is known with certainty, the conductance was estimated by calibration to obtain the best match with available stream flows.

Whenever possible, a whole island was simulated as a single unit to avoid the difficulties in setting up internal boundaries between subunits. In such a case, only the ocean boundary is needed for the assessment process. In other cases, limitations regarding grid sizes and the number of cells necessitated dividing the island into parts; for example Maui was divided into east and west sections. In this case, the division was done by using sector and system boundaries as defined by Department of Land and Natural Resources (DLNR). Appropriate hydrologic properties were assigned to the contacts between systems or sectors, depending on the physical nature of the boundary. For example, a no-flow condition is assigned if the predominant groundwater flow direction is oriented parallel to a boundary. In other instances, the model grid was extended across these regulatory boundaries to allow simulation of leakance between adjacent aquifer systems.

Aquifer thickness was estimated based on aquifer's geological characteristics and location, and the position of saltwater–freshwater interface. The Ghyben-Herzberg equation was used in estimating the bottom boundary for freshwater-lens aquifers. For high-level waters, such a principle is generally not valid because an effective aquifer thickness would be much smaller than that estimated by the Ghyben-Herzberg relationship. The Ghyben-Herzberg principle assumes that groundwater flow is mainly horizontal. However, as the water table transitions from high-level groundwater to freshwater-lens groundwater, the horizontal flow assumption of the Ghyben-Herzberg principle is no longer valid (Izuka and Gingerich 1998). Furthermore, compaction of the rock at deeper depths due to the weight of the overburden greatly reduces the permeability (Oki 1997). Hence, the depth of the midpoint of the freshwater–saltwater transition zone is difficult to estimate and currently no information exists about the maximum effective depth at which significant groundwater flow occurs in dike-confined aquifers. An arbitrary value of 1,000 m below sea level was assumed. To prevent a sharp truncation at −1,000 m, a gradual increase was applied when computing the bottom boundary to all groundwater elevations greater than 6.25 m, for which the bottom boundary equals −250 m (Fig. 2). These values are random and new studies are needed to better define an effective aquifer thickness for sharp interface models that ignore density effects.

The use of MODFLOW facilitated modeling of infiltration galleries, or Maui-type shafts, which typically consist of a series of infiltration tunnels excavated near sea level, pumping water from a central shallow sump (Watson 1964). Due to partial penetration of infiltration tunnels and high pumping rates associated with these sources, the groundwater flow pattern near the tunnel is three-dimensional in nature.

The amount of well discharge is critical in determining the size and shape of the simulated assessment areas. For wells located in State Water Management Areas, as designated by DLNR, the maximum allocated daily pumping rate was used. For wells outside these selected areas, the pump capacity of the well was used unless the

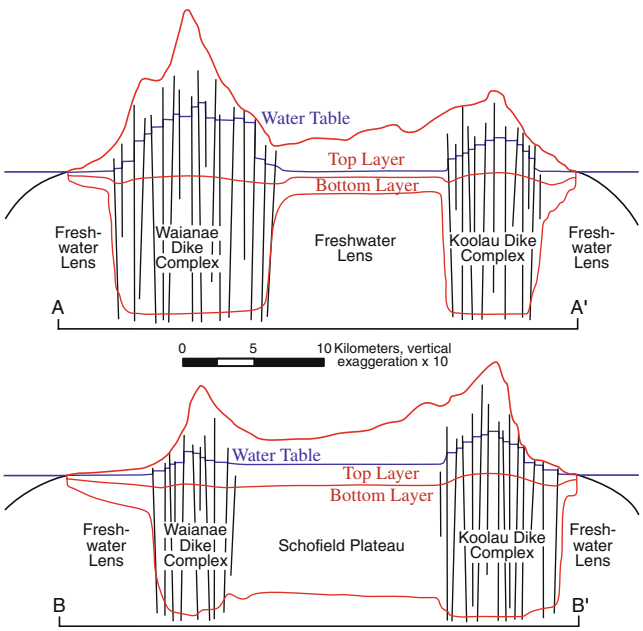


Fig. 2 Schematic cross-section representation of the 2-layered MODFLOW model grid (Island of Oahu), and conceptual illustration of the water table, interface between top and bottom layer, and calculated bottom boundary

DOI 10.1007/s10040-009-0548-6

sustainable yield of the entire aquifer sector was exceeded. In this case, the actual withdrawal was proportionally increased until the sustainable yield value was reached. The withdrawal of all Oahu aquifers at the time of the assessment was less than the CWRM established sustainable yield (as per the 1991 Water Resources Protection Plan). Wells falling within one model cell were treated as a single well, and the total withdrawal from these wells was used as input to the model. Furthermore, wells allocated as "batteries" by DLNR were treated as a single pumping center.

General guidelines for identifying values for aquifer parameters were based on available modeling data, especially those modeling studies done by the USGS's Pacific Island Water Science Center, Honolulu, Hawaii (Oki 1997, 1998, 2002; Izuka and Oki 2002). The values of these hydrologic parameters were estimated by calibration and greatly influence the size and shape of the delineated assessment areas, and certainly cause uncertainties in the final assessment results. The calibration process included matching available hydraulic head values with those simulated by MODFLOW. An automatic fitting procedure was used in the calibration process through the program PEST (Doherty 2004). Data input to PEST includes initial value of parameters and their range of plausible values. In some cases, a manual method was adopted including numerous runs to obtain the best fit possible. Hydraulic-conductivity values were constrained within the known ranges available in the literature (Hunt 1996) and match recently assessed values (Rotzoll 2007; Rotzoll and El-Kadi 2008). Hydraulic conductivity values of the calibrated model for flank lavas range from 70 to 700 m/d, although the majority was modeled using 500 m/d. The hydraulic conductivity assigned to caprock sediments varied between 30 and 60 m/d and those for the dike complexes are between 0.01 and 1.5 m/d. Values of porosity of Hawaii lavas cover a wide range, from less than 0.05 to 0.5 (Peterson and Seghal 1974; Lau and Mink 2006). To be on the conservative side, the SWAP Advisory Committee selected a porosity of 0.05 to be used for the SWAP assessments, which would result in a relatively large assessment area.

Following a satisfactory calibration of MODFLOW, MODPATH backwards particle-tracking simulations were run to determine the area that contributed water to wells within the previously stated times of travel. Finally, a capture-zone delineation (CZD) polygon was traced around the outside tracks and termination points of the particle traces. Uncertainties in the CZDs estimates were evaluated by using sensitivity analyses regarding values of some key parameters, and regarding overlooking the hydrodynamic dispersion process. The calculations were completed for the Island of Molokai. Six sensitivity simulations were run to test the effect of varying hydraulic conductivity, recharge, and position of the boundary between the dike zone and the flank lavas.

In two sensitivity runs, values of the hydraulic conductivity of the formations were increased and decreased by 50% relative to the base case. The resulting changes were minor with a little change to the size of the CZD. For the types of boundary conditions utilized, which included constant flux, the particle velocity was not drastically affected. Recharge was varied by ±25%, but no major changes occurred for the size of the CZD for the case of 25-year travel time. Change in the length of the 2-year CZD was about 10% as compared to the base case.

More significant changes resulted for the 25-year CZD when porosity was varied. The model porosity was doubled to 0.1, resulting in about a 5% reduction in the length of the CZD zone. Decreasing the porosity to 0.025 resulted in a slight increase in such a length. The sensitivity of the model to changes in porosity was limited by the particles encountering the low permeability of the dike zone. The 2-year CZD as expected experienced a change in length inversely proportional to the change in porosity.

Changing the position of the dike zone/flank lava boundary resulted in changes in the CZD length, but more importantly changed the trajectory of the particle tracks. This is consistent with the oblique effect on groundwater flow direction as particles cross a hydraulic conductivity contrast interface at an angle, which causes a deflection toward the area of higher conductivity (Freeze and Cherry 1979). The envelope of various sensitivity runs is compared to the original runs (Fig. 3). The results are for the base case 25-year CZD overlain on the 2-, 10-, and 25-year sensitivity composite CZDs. The increase in CZD width of the 25-year time of travel zone is the result of the shift in particle track caused by the repositioning of the flank lava/dike zone boundary.

Overlooking dispersion underestimates the size of the CZD and can result in contamination outside of a CZD being captured by a drinking water well. To assess the influence of dispersion on the size of the CZD, a source area with a chloride concentration of 100 mg/L was simulated at the upper end of the 25-year CZD. A MT3D simulation was run for 25 years. A dispersivity of 35 m was used based on analysis of rock cores in central Oahu (TEC Inc. 2007). The resulting chloride plume is overlain on original and the enhanced CZDs (Fig. 3).

Potential contaminating activities

A potential contaminating activity (PCA) is defined as a facility or activity that stores, transmits, uses, or produces contaminants, and has the potential to release contaminants that may influence water quality. PCAs that reside within the assessment area of a well were inventoried to primarily provide more information about land-use activities surrounding the source. It is also the initial step to determine susceptibility of the well to contamination. A generic list of PCAs and their associated contaminants were created and served as the basis for the inventory for all sources. Inventories categorized PCAs as having very high, high, medium, and low potential for contamination (Table 1). Factors that were considered when placing a PCA into a certain category included the nature of the activity, relevant contaminants, and association with

 DOI 10.1007/s10040-009-0548-6

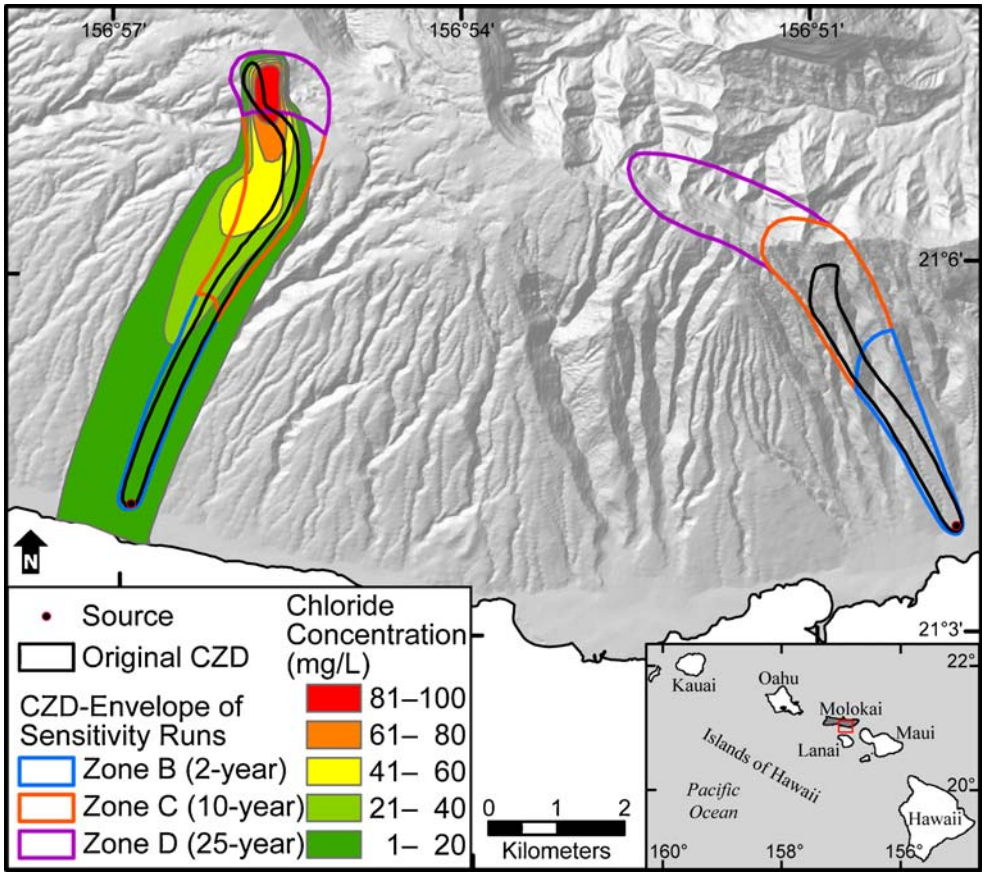


Fig. 3 Capture-zone envelope of various sensitivity runs compared to the original runs on the Island of Molokai, and simulated chloride plume of initial concentration of 100 mg/L started at the upper end of the 25-year CZD

historical incidents of contamination. To simplify the analysis, site-specific considerations such as the existence and implementation of best-management practices, were not considered in the ranking of PCAs.

The first step in the inventory was identifying available sources of PCA information, which included published and unpublished reports, federal and state's databases, and GIS coverages. Data sources such as land use maps, were processed into GIS, which also helped in exploring and analyzing PCA data by revealing hidden patterns, relationships, and trends that were not readily apparent in spreadsheets or statistics. Government database queries and interviews with local and state government officials also provided useful PCA data.

In addition, a number of PCAs were identified through an electronic yellow pages search. PCAs were searched and identified by associated standard, industrial classification numbers. The ArcGIS geocoding feature was used to tie the addresses of the facilities identified in the yellow pages program to actual locations on a map. Most of the base map components and island infrastructure were acquired from the state of Hawaii GIS website (State of Hawaii 2007).

All PCAs that fell within a delineated assessment area were printed on a map and identified in an associated table. Additional state and local data sources that were not processed into GIS were also used to supplement the inventory. The initial inventory was limited to those PCAs identified in the PCA list as of very high, high, and medium potential. The "low" category for PCAs was not inventoried at this stage. However, the presence of low PCAs within the assessment area was documented.

Finally, the information identified in the initial inventory of PCA maps and tables were verified during field surveys. PCAs that did not have adequate address information were then surveyed by GPS. The surveys also identified and located any PCAs that were not included in the initial inventory.

Source water susceptibility

US EPA Guidance defines "susceptibility" as the potential for a public water system to draw water contaminated by inventoried PCAs at concentrations that would pose concern (US EPA 2006). In this study, susceptibility determination incorporated the PCA contamination potential, distance of the PCA from the well, area occupied by the PCA or number of PCAs within the assessment zone, aquifer sensitivity, and whether a contaminant was detected in the source water. Contamination potential was based on the rank of each PCA as medium, high, or very high. In general, the closer a PCA is to the well, the higher the likelihood that a contaminant released at the site will adversely impact the water pumped from the well.

 DOI 10.1007/s10040-009-0548-6

Table 1 Listing of data sources and scoring for potential contaminating activities

PCA name	PCA type	GIS data type	Data source	PCA score
Resource Conservation and Recovery Act (RCRA), National Corrective Action Prioritization System sites	Very high	Point	EPA GIS database	20
Airport, maintenance and fueling areas	Very high	Point	TMK (tax map key) GIS data	10
Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) sites	Very high	Point	Superfund sites/EPA Websites	10
Cesspools	Very high	Polygons	HDOH cesspool records and TMK data	Table 2
Chemical/petroleum processing/storage	Very high	Arc	Map review	10
Confined animal feedlot	Very high	Point	Personal communication with US Department of Agriculture (USDA)	10
Dry cleaners/processing	Very high	Point	Telephone book	10
Gas stations	Very high	Point	County GIS Databases and telephone book	10
Injection wells/dry wells/ sumps	Very high	Point	HDOH database	10
Illegal activities/ unauthorized dumping	Very high	Point	Located during field survey	10
Improperly abandoned wells	Very high	Point	Source of data could not be located	10
Large quantity hazardous waste generators	Very high	Point	EPA and HDOH databases	10
Landfills/dumps/or historic dumps	Very high	Point	County records	10
Leaking underground storage tank (UST)	Very high	Point	HDOH database	10
Metal plating/finishing/fabricating	Very high	Point	Telephone book	10
Military installations	Very high	Polygon	Map review	10
Other crops using soil fumigants (direct application into soil)	Very high	Polygon	USDA agricultural land-use database	10
Plastics/synthetic fabricators	Very high	Point	Telephone book	10
Pesticides or herbicides mixing and loading sites	Very high	Point	Personal communication with USDA	10
Pineapple cultivation	Very high	Polygon	USDA agricultural landuse database	Table 2
Power plants	Very high	Point	Data collected from map review and TMK GIS database	10
RCRA, treatment, storage or disposal facility	Very high	Polygon	HDOH database	10
Recorded spills	Very high	Polygon	HDOH database	10
Sugarcane cultivation	Very high	Polygon	USDA agricultural landuse database	Table 2
Underground injection of commercial/industrial discharge	Very high	Point	HDOH database	10
Wastewater treatment plants	Very high	Point	HDOH database	10
Wood treatment facilities	Very high	Point	Telephone book	10
Auto body shops	High	Point	County records	6
Automobile repair shops	High	Point	County records	6
Boat services/repair/refinishing	High	Point	Telephone book	6
Construction or farm machinery repair/maintenance	High	Point	Source of data could not be located	6
Diversified agriculture	High	Polygon	USDA agricultural landuse database	Table 2
Fleet/trucking/bus terminals	High	Point	Telephone book	6
Furniture repair or manufacturing	High	Point	Telephone book	6
Golf courses	High	Polygon	County records	6
Junk yards/scrap/salvage yards	High	Point	County records	6
Lagoons/liquid wastes	High	Point	Not located within a CZD	6
Machine shops	High	Point	Telephone book	6
Photo processing/printing	High	Point	Telephone book	6
Pesticide distributors/professional applicators	High	Point	Telephone book	6
Research laboratories	High	Point	Telephone book	6
Reclaimed water R2	High	Polygon	County records	6
Sewer lines	High	Line	County records	Table 2
Septic systems	High	Polygon	DOH septic system database	Table 2
RCRA, small quantity generators	High	Point	EPA and HDOH RCRA database	6
UST non-leaky, or no leaks reported	High	Point	HDOH UST database	6
Utility stations/maintenance areas	High	Point	County records	6
UST- non-regulated, not yet upgraded or registered	High	Point	Data not available	6
Wells geothermal, production and injection	High	Point	DOH database	6
Above ground storage tanks	Medium	Point	County records	3
Artificial recharge projects (non-potable water)	Medium	Polygon	During survey such projects were not found in any CZD	3
Campgrounds	Medium	Polygon	DLNR and county records	3
Car washes	Medium	Point	County GIS data and telephone book	3
Cement/concrete plants	Medium	Point	Telephone book	3

 DOI 10.1007/s10040-009-0548-6

Table 1 (continued)

PCA name	PCA type	GIS data type	Data source	PCA score
Cemetery	Medium	Point	County records	3
Contractor or government agency equipment storage yard	Medium	Point	County records	3
Food processing (sugarcane mills)	Medium	Point	Personal communication with USDA	3
Hardware/lumber/part stores	Medium	Point	Telephone book	3
High schools	Medium	Point	County GIS data	3
Hospitals	Medium	Point	County permit records	3
Parking lots	Medium	Polygon	County permit records	3
Parks	Medium	Polygon	DLNR and county records	3
Reclaimed wastewater irrigation - R1 water	Medium	Polygon	HDOH and county records	3
Residential parcels	Medium	Point	TMK database	Table 2
Transportation corridors	Medium	Line	State GIS database	3
RCRA, conditionally exempt generator	Medium	Point	RCRA database	3
Storm drain discharge points, detention facilities and dry wells	Medium	Point	County records	3
Waste transfer/recycling stations	Medium	Point	County records	3

Hence, PCAs falling in zone A have a higher score than similar ones falling in other zones. Since zone A is intended to recognize the potential for direct spills, the type of well construction was also considered.

A PCA was classified as either a point or a non-point source. Non-point sources included pineapple, sugarcane, diversified agriculture, field crops, residential parcels, and sewer lines. All other sources were considered as point sources. Cesspools and septic systems were considered as non-point sources because the numbers of cesspools found in a CZD were so high that the score would be out of bounds. Naturally, the area over which an activity is occurring will influence its susceptibility. For example, the potential for contamination is high for large impacted spatial areas such as the case for agricultural lands. The scoring was also based on the density of the activity within the delineated source area for cesspools, residential parcels, septic systems, and sewer lines.

Aquifer sensitivity takes into account the vulnerability of the aquifer based on the hydrogeologic setting of the aquifer as defined by Mink and Lau (1990). Among other factors, aquifers were classified according to vulnerability to contamination. The aquifer sensitivity attribute in the current study was based upon such vulnerability, and was rated as high, moderate, and low and is reported for each source in the final report. Current research on the environmental/health risk posed by on-site sewage disposal systems is incorporating unsaturated zone characteristics such as depth to groundwater, soil thicknesses less than 1.5 m, soil permeability, and filtration capacity. The results of this study will be considered in future revisions to the SWAP methodology.

Finally, if a contaminant commonly associated with a PCA is detected in the well, then that particular PCA should receive a higher susceptibility rating. The Hawaii Department of Health (HDOH) currently monitors water sources for a number of constituents (Hawaii Department of Health 2006). If contamination was detected, scoring for the groundwater source was increased by a fixed number of points.

A numerical scoring system was used to rank the relative susceptibility of the drinking water source to each PCA (Tables 1 and 2, Fig. 4). Values were assigned for each of the criterion already described and the susceptibility score for a particular PCA was the sum of the scores for all criteria. The higher the score, the higher the potential is for contamination from that particular PCA. The numerical score serves as a basis for comparison among PCAs, water sources, and water systems. A summary table was generated for each source, which listed all the PCAs in the assessment area in order of importance (i.e., highest to lowest score). In addition, a cumulative susceptibility score for the well was provided. The cumulative susceptibility score is the sum of all the PCA scores.

Example application

The above approach was applied to all groundwater sources of the state of Hawaii (Whittier et al. 2004a, b, c, 2005a, b, 2006b). As an example, the application to the groundwater sources on the Island of Oahu is described in the following. Various hydrogeologic zones characterize the Island of Oahu (Fig. 1). Detailed description of these zones and ranges of their hydraulic conductivities can be found in Hunt (1996) or Oki (1998). The model grid used to delineate the CZDs consisted of 283,860 cells (243,208 active cells) arranged in 285 rows, 498 columns, and two layers. Lavas in most areas were assumed to be vertically homogeneous due to the absence of information characterizing their vertical layering; however, near the coast, inclination of the basalt/caprock interface was reflected in the layering.

Table 2 Scoring algorithms for non-point potential contaminating activities

PCA type	Score type	Score
Pineapple cultivation	Area (ha)	0.08 × area + 6
Sugarcane cultivation	Area (ha)	0.08 × area + 6
Diversified agriculture	Area (ha)	0.05 × area + 3
Other field crops	Area (ha)	0.05 × area + 3
Cesspool	Count	0.250 × count + 4
Septic system	Count	0.125 × count + 2
Sewered parcel	Count	0.005 × count + 4
Residential parcel	Count	0.0025 × count + 2

 DOI 10.1007/s10040-009-0548-6

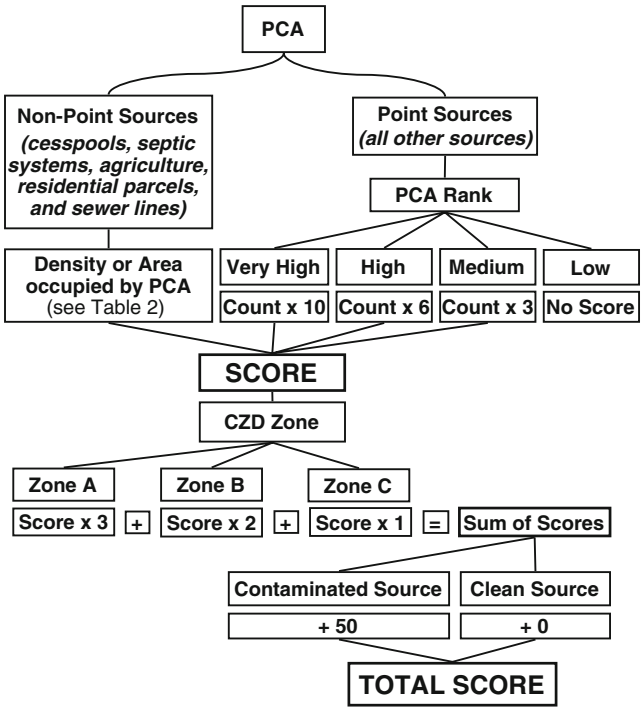


Fig. 4 Numerical scoring system to rank the relative susceptibility of the drinking water sources to potential contamination activities

The upper model boundary coincided with the simulated water table considering that MODFLOW ignores flow in the unsaturated zone (Fig. 2). Care was taken, however, in avoiding wet and dry model cells based on ground-surface topography. The bottom boundary of the modeled area was based on an assumed sharp saltwater–freshwater interface. The lateral boundaries included specified head at the shoreline. Assigned values varied between 0.0 to 2.2 meters above mean sea level (m amsl), based on available data (Oki 1998), with the largest value applied where the southern Oahu caprock is the thickest. A more accurate setup of the boundary condition would include seepage through the caprock (Oki 1998). A simplified approach was used by implementing a specified head condition with values obtained from available records. In this manner, the size of the aquifer model was drastically reduced by eliminating areas extended to the ocean side. Such an approximation is justified considering that drinking-water wells are far from the shoreline and the results are not likely to be seriously impacted.

For the calibration, withdrawal from sources was based on average rates from 1995. Recharge, the only source of water into the model, was based on a water budget for Oahu using a 1985 land-use scenario (Shade and Nichols 1996). This water budget was comprehensive, accounting for the contributions of rainfall, surface-water runoff, potential evapotranspiration, and irrigation water return. The assigned recharge values varied from 0.0001 m/d in the coastal leeward areas to 0.011 m/d at the summit area of the Koolau Range (Fig. 5). In the calibration process, model parameters were adjusted to minimize the error in hydraulic head estimates. Both measured and modeled hydraulic-head values found in scientific literature were used in the comparison. Measured stream flows were also considered in the calibration process and the objective was to minimize the difference between estimated and simu-

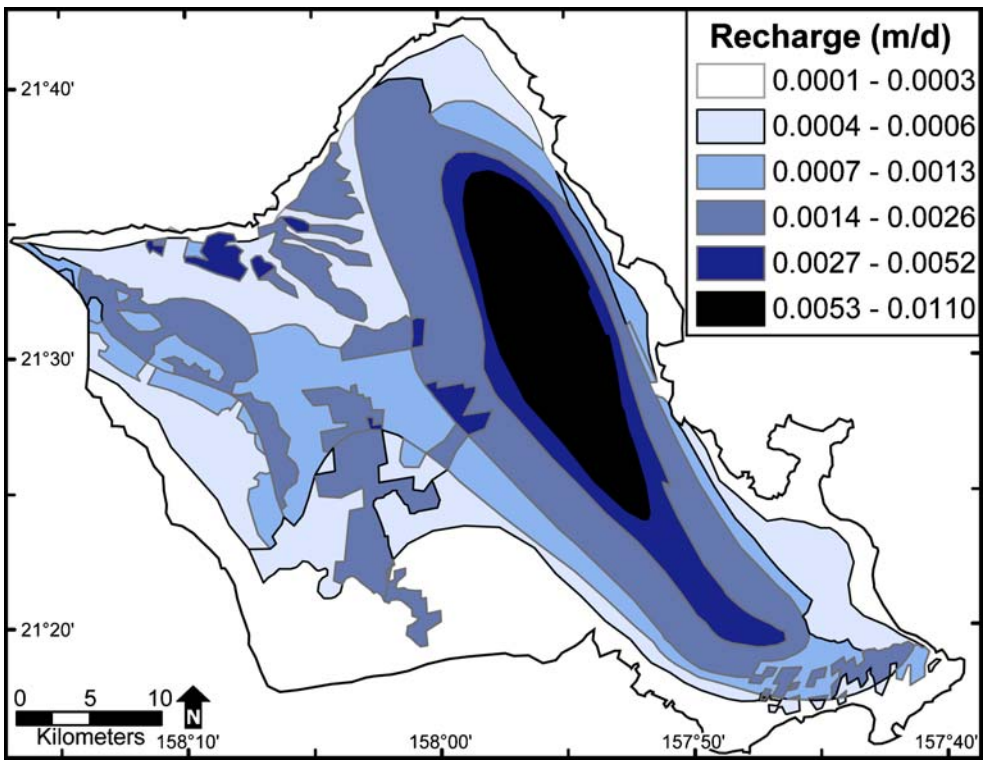


Fig. 5 Distribution of recharge values used in the model for Oahu, Hawaii

 DOI 10.1007/s10040-009-0548-6

lated values of stream baseflow. Parameters adjusted were aquifer hydraulic conductivity, stream arc conductance, and HFBs conductance. The hydraulic characteristic of arcs representing volcanic dikes and faults were also included in the calibration process.

Calibration points were divided into two different categories depending on whether they fell within the freshwater-lens groundwater or in the high-level groundwater areas. For the freshwater-lens water areas, simulated water levels were calibrated to the average water level for each well recorded in the USGS Water Year 1999 Report (Hill et al. 2000). This was augmented by water levels contained in the well database of the state's Commission on Water Resources Management (CWRM). However, priority was given to USGS data because all of these water levels were recorded during the same period. The water levels contained in the CWRM well database included water levels taken when the wells were first drilled as well as periodic measurements following well installation. The groundwater elevations calculated by this model were also compared to those calculated by Oki's (1998) central Oahu groundwater model using the 1995 allocated pumping rates.

Measured water levels used for the high-water areas were supplemented with the generalized water-table elevations in Nichols et al. (1996). In addition, spring and tunnel elevations, recorded in the CWRM well database, represent the minimum possible water level. The model was calibrated such that the simulated water levels equaled or exceeded these elevations. Following the calibration process, the CZDs were delineated. Pumping rates used in the delineation process were based on those allocated by the CWRM (2001).

Results and discussion

Simulated groundwater elevations for the Island of Oahu reach 6 m in freshwater-lens aquifers and decrease towards the coast (Fig. 6). The heads in the high-level systems range from 50 m (Schofield plateau) to over 400 m amsl (dike complexes). The final calibration for the freshwater-lens and high-level aquifers is shown comparing measured and simulated heads. A close match was obtained for the freshwater-lens water areas with a mean error of 0.16 m, an RMS error of less than 0.58 m, and a normalized RMS error of 0.10. The largest error occurred in the Mokuleia groundwater area, on the western end of the North Shore aquifers, where the simulated groundwater elevation was 1 m higher than the measured groundwater elevation. The calibration in high-level systems was also acceptable with a normalized RMS error of 0.12, considering that most of the sources are located in freshwater-lens aquifers.

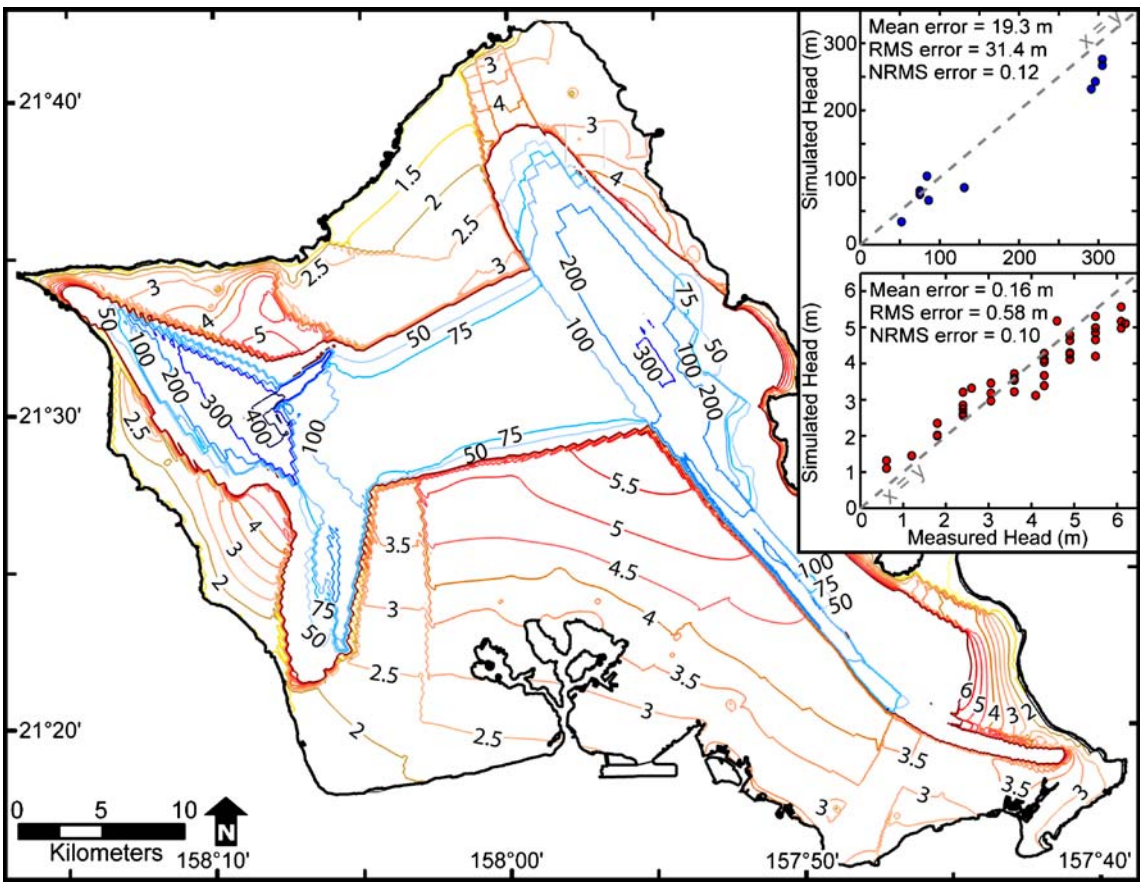


Fig. 6 Calculated groundwater elevations in meters above sea level and calibration plot for high level (*blue*) and freshwater-lens aquifers (*red*), Oahu, Hawaii. The *mean error*, *RMS error*, and normalized RMS error (*NRMS*) are indicated in the small plots

 DOI 10.1007/s10040-009-0548-6

The CZDs for Oahu were delineated after satisfactory calibration of the groundwater flow model (Fig. 7). The density of the wells in the Pearl Harbor Aquifer system made individual zones impractical due to the overlapping particle tracks and the influence that withdrawal in each well had on neighboring wells. In the Pearl Harbor and Honolulu Aquifer Systems the CZDs cover most of the land area between the caprock, and the south Schofield Dam and the Koolau Rift Zone. Elsewhere on Oahu the well density was lower, resulting in fewer combined CZDs, and with the exception of the North Shore Aquifer System, the CZDs were much smaller because of the lower hydraulic-conductivity values and the closer proximity of the wells to rift zones.

Unexpectedly, the length of many 2-year CZDs was about two thirds of the length of the 10-year CZDs. Two factors account for this. First, as the tracking particles encounter the low hydraulic conductivity of the high-level water areas, the groundwater velocity decreases significantly. Second, the multi-layer approach and the three-dimensional capability of the MODFLOW model allowed tracking particles to terminate at the water surface earlier than the designated time of travel.

PCA scores for various groundwater sources are based on the susceptibility analysis (Fig. 7). Areas of highest susceptibility are located, and as should be expected, in CZDs dominated by high-density population centers, and by agricultural, industrial, and military areas (Pearl Harbor and Honolulu). It has to be stressed again that the numerical scores serve only as a basis for comparison among water sources. A need exists to reevaluate and modify the methodology by considering formal risk methods.

Assumptions and limitations

Limitations of this study are mostly related to accuracy of the numerical groundwater models used in delineating SWAP zones B and C. Inaccuracies are related to the nature of the models themselves concerning processes simulated as well as the lack of complete and accurate data for Hawaii aquifers. The main simplifying assumptions in the models include neglecting flow in the unsaturated zone, chemical reactions, and chemical dispersion. Overlooking travel time in the unsaturated zone overestimates contamination risk, i.e., more conservative, by assuming that a contaminant would immediately reach the water table. Such a travel time can be significant, on the order of 15 years in some areas where the water table lies several hundred meters below the ground surface (Orr and Lau 1988). The approach is also conservative by ignoring potential degradation and other chemical reactions that likely reduce the concentration of certain non-conservative chemicals. However, there is an expected error due to ignoring the dispersion process resulting in a decrease in the expected size of the zone. Nevertheless, overlooking dispersion is justified considering that uncertainties in estimating the dispersion parameters would overshadow any expected success in delineation of the assessment zone.

Hawaii hydrogeology is complex where aquifers are composed of basalts, geologic barriers, volcanic rift zones

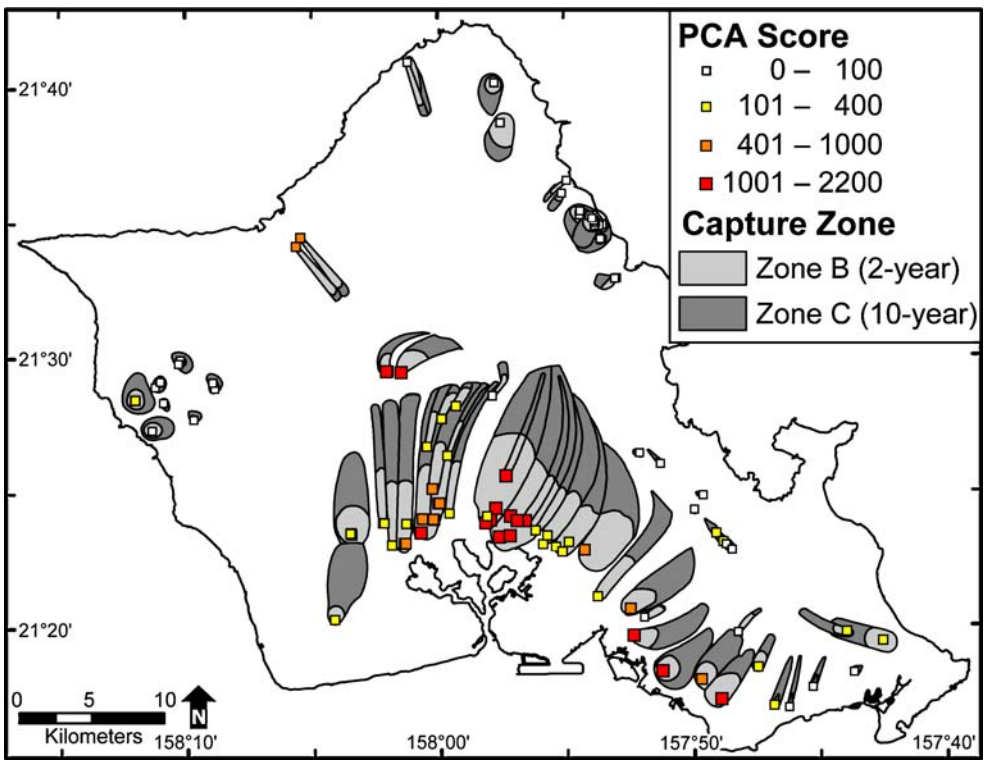


Fig. 7 Delineated capture zones for Oahu's groundwater sources and potential-contaminating-activities scores based on the susceptibility analysis

 DOI 10.1007/s10040-009-0548-6

containing intersecting low-permeability vertical dikes, and valleys filled with sediment that act as barriers to flow through layered basalts. Research to characterize aquifer heterogeneity on multiple scales is non-existent. Hence, conventional calibration was used in estimating effective values over aquifer systems. A steady-state model was applied that ignored temporal variability in recharge and pumpage values, which would cause variations in the well capture zones at various time scales.

More studies are needed to improve on modeling the rift zones and valley fills, which, and as discussed earlier, were modeled by using HFBs. Such an approach greatly improved the simulated results by capturing the anisotropy of the rift zones and preventing the overestimation of well drawdowns. However, the number and placement of the HFBs were somewhat arbitrary and work is in progress to formalize this approach.

The study included creating an inventory of potential contaminant sources. The inventory is a snapshot in time and may include inaccuracies. For example, included in these inventories were HDOH and EPA hazardous waste contamination databases (i.e., leaking underground storage tanks, Resource Conservation and Recovery Act sites, and Comprehensive Environmental Response, Compensation and Liability Act sites.). Such lists may include sites that are no longer active.

Future update and utilization

Models based on SWAP have been used for other studies, including developing a numerical groundwater flow model around the Red Hill Ridge on the Island of Oahu (Rotzoll and El-Kadi 2007). The existing regional SWAP model of the Island of Oahu was used to provide the boundary conditions for a localized groundwater flow model for the Red Hill area. Calibrated parameters were used in a subsequent study that dealt with a contaminant fate and transport model (Whittier 2007). A future study will utilize SWAP models in establishing an inspection protocol for on-site sewage disposal systems. Maps for the depth to water table developed by SWAP have been used by the state of Hawaii's Department of Agriculture to assess potential contamination by pesticides. This study has been included as an integral part of the proposed Hawaii State's Water Protection Plan. The results are also used in updating aquifer sustainable yields, which are used in water use decisions.

The SWAP will be maintained and improved as new information becomes available. In addition to conducting assessments for new sources, future efforts will include developing a GIS database management system for SWAP. The system will integrate into the GIS various databases such as PCA data sheets, aquifer data, and well production data. In addition, tools for PCA and water quality parameters searching and summary reporting will be designed. Finally, documents such as photographs and scanned drawings will be linked to the GIS system. The scoring system needs to be reevaluated and modified by enhancing the susceptibility analyses by considering formal risk methods.

Implementing SWAP is beyond the scope of this study. However, the program will benefit future protection plans, proposed by federal, state, and local agencies, by providing useful data, technical resources, and communication networks. Already various water purveyors are adopting SWAP results for developing protection plans for their respective water sources. The Island of Maui has achieved a greater progress, which benefited from this study. The effectiveness of source water protection can be greatly increased and enhanced through integration of SWAP with other programs, which oversee activities that may affect water quality, such as agriculture, forestry, transportation, fisheries, wildlife, land use, hazardous waste, solid waste, recreation, parks and public lands. Coordination and collaboration efforts are crucial to the success of SWAP in sharing existing resources, programs and data, and reducing duplication of efforts.

Acknowledgements The research was funded by the Hawaii Department of Health. This paper is published as a part of the continuing University of Hawai'i at Mānoa Water Resources Research Center "Contributed Papers" series. As such it will be assigned a sequential "WRRC-CP" number upon publication. This number, along with the abstract of the paper, will then be posted at the WRRC website http://www.wrrc.hawaii.edu/index.html.

References

California Department of Public Health (2007) Drinking Water Source Assessment and Protection (DWSAP) Program. http://www.cdph.ca.gov/certlic/drinkingwater/Pages/DWSAP.aspx. Cited 6 Feb 2008

Commission on Water Resource Management (2001) Land Water Use Permit Index. CWRM, Department of Land and Natural Resources, State of Hawaii, Honolulu

Doherty JL (2004) PEST: model-independent parameter estimation, user manual, 5th edn. Watermark Numerical Computing, Brisbane, Australia

EMS-i (2006) Groundwater Modeling System 6.0. Environmental Modeling Systems, Inc., South Jordan, UT. http://www.ems-i.com/GMS/GMS_Overview/gms_overview.html. Cited 4 Oct 2006

Freeze RA, Cherry JA (1979) Groundwater. Prentice Hall, Upper Saddle River, NJ

Gingerich SB, Oki DS (2000) Ground water in Hawaii. US Geol Surv Fact Sheet 126–00

Harbaugh AW, Banta ER, Hill MC, McDonald MG (2000) MODFLOW-2000, the US Geological Survey modular ground-water model: user guide to modularization concepts and the ground-water flow process. US Geol Surv Open-File Rep 00–92

Hawaii Department of Health (2006) Contaminants to be tested in all new sources of drinking water. HDOH, Safe Drinking Water Branch. http://www.hawaii.gov/health/environmental/water/sdwb/index.html. Cited 11 Nov 2008

Hill BR, Fontaine RA, Taogoshi RI, Teeters PC (2000) Water resources data, Hawaii and other Pacific areas, Water year 1999, vol 1. US Geol Surv Hawaii Water-Data Rep HI-99–1, US Geological Survey Hawaii, Honolulu

Horsely and Witten Inc (1992) Groundwater hydrology, contamination and management. Final Report on the Demonstration Project conducted on Molokai, Hawaii, Horsely and Witten, Sandwich, MA

Hunt CD (1996) Geohydrology of the Island of Oahu, Hawaii. US Geol Surv Prof Pap 1412-B

Izuka SK, Gingerich SB (1998) Estimation of the depth to the freshwater/salt-water interface from vertical head gradients in wells in coastal and island aquifers. Hydrogeol J 6(3):365–373

Izuka SK, Oki DS (2002) Numerical simulation of ground-water withdrawals in the Southern Lihue basin, Kauai, Hawaii. US Geol Surv Water Resour Invest Rep 01-4200

Lau LS, Mink JF (2006) Hydrology of the Hawaiian Islands. University of Hawaii Press, Honolulu

Macdonald GA, Abbott AT, Peterson FL (1983) Volcanoes in the sea: the geology of Hawaii, 2nd edn. University of Hawaii Press, Honolulu

Mink JF, Lau LS (1990) Aquifer identification and classification for Oahu: groundwater protection strategy for Hawaii. Water Resources Research Center Technical Report 179, University of Hawaii, Honolulu

New Jersey Department of Environmental Protection (2005) Source Water Assessment Program, http://www.nj.gov/dep/swap/. Cited 5 Feb 2008

Nichols WD, Shade PJ, Hunt CD (1996) Summary of the Oahu, Hawaii, regional aquifer-system analysis. US Geol Surv Prof Pap 1412-A

Oki DS (1997) Geohydrology and numerical simulation of the ground-water flow system of Molokai, Hawaii. US Geol Surv Water Resour Invest Rep 97–4176

Oki DS (1998) Geohydrology of the central Oahu, Hawaii, ground-water flow system and numerical simulation of the effects of additional pumping. US Geol Surv Water Resour Invest Rep 97–4276

Oki DS (2002) Reassessment of ground-water recharge and simulated ground-water availability for the Hawi area of north Kohala, Hawaii. US Geol Surv Water Resour Invest Rep 02–4006

Oki DS (2005) Numerical simulation of the effects of low-permeability valley-fill barriers and the redistribution of ground-water withdrawals in the Pearl Harbor area, Oahu, Hawaii. US Geol Surv Sci Invest Rep 2005–5253

Oki DS, Lau LS, Mink JF (1992) Wellhead protection methodology for Hawaii. Water Resources Research Center Special Report 01, University of Hawaii, Honolulu

Orr S, Lau LS (1988) Trace organic (DBCP) transport simulation of Peal Harbor aquifer, Oahu Hawaii: method of characteristics, phase II. Water Resources Research Center Technical Report 175, University of Hawaii, Honolulu

Peterson FL, Seghal MM (1974) Determining porosity with neutron logs from Hawaiian basaltic aquifers. Technical Report 80, Water Resources Research Center, University of Hawaii, Honolulu

Pollock DW (1994) User's guide for MODPATH/MODPATH-PLOT, version 3: a particle tracking post-processing package for MODFLOW, the US Geological Survey finite-difference ground-water flow model. US Geol Surv Open-File Rep 94–464

Rotzoll K (2007) Hydraulic parameter estimation using aquifer tests, specific capacity, ocean tides, and wave setup for Hawaii aquifers. PhD Thesis, University of Hawaii, Honolulu

Rotzoll K, El-Kadi AI (2007) Numerical ground-water flow simulation for Red Hill fuel storage facilities, NAVFAC Pacific, Oahu, Hawaii. Prepared for TEC Inc., University of Hawaii, Water Resources Research Center, Honolulu

Rotzoll K, El-Kadi AI (2008) Estimating hydraulic conductivity from specific capacity for Hawaii aquifers, USA. Hydrogeol J 16(5):969–979

Shade PJ, Nichols WD (1996) Water budget and effects of land-use changes on ground-water recharge, Oahu, Hawaii. US Geol Surv Prof Pap 1412-C

State of Hawaii (2007) Hawaii Statewide GIS Program. Office of Planning, State of Hawaii, Honolulu. http://hawaii.gov/dbedt/gis/. 5 Feb 2008

TEC Inc (2007) Red Hill Bulk Fuel Storage Facility final technical report, Pearl Harbor, Hawaii. Prepared for the Dept. of the Navy, Naval Facilities Engineering Command, Pacific Contract NO. N62742–02-D-1802, CTO 007, TEC Inc, Honolulu

US EPA (2006) State source water assessment and protection programs guidance. US EPA, Washington, DC. http://cfpub.epa.gov/safewater/sourcewater/sourcewater.cfm?action=Assessments. 28 Jan 2007

Virginia Department of Health (2007) Source Water Assessment Program (SWAP). Virginia Department of Health, Richmond, VA. http://www.vdh.virginia.gov/DrinkingWater/source/swap.htm. 2 Feb 2008

Watson LJ (1964) Development of ground water in Hawaii. J Hydraul Div ASCE 90(HY6):185–202

Whittier RB (2007) Red Hill Fuel Storage Facility final contaminant transport simulations using numerical models for tier 3 risk evaluation, Pearl Harbor, Hawaii. Prepared for the US NAVY, NAVFAC Pacific, Oahu, HI

Whittier RB, Rotzoll K, Dhal S, El-Kadi AI, Ray C, Chen G, Chang D (2004a) Hawaii Source Water Assessment Program report, vol 3: Island of Kauai. Prepared for Hawaii Dept. of Health, Water Resources Research Center, University of Hawaii, Honolulu

Whittier RB, Rotzoll K, Dhal S, El-Kadi AI, Ray C, Chen G, Chang D (2004b) Hawaii Source Water Assessment Program report, vol 4: Island of Lanai. Prepared for Hawaii Dept. of Health, Water Resources Research Center, University of Hawaii, Honolulu

Whittier RB, Rotzoll K, Dhal S, El-Kadi AI, Ray C, Chen G, Chang D (2004c) Hawaii Source Water Assessment Program report, vol 7: Island of Oahu. Prepared for Hawaii Dept. of Health, Water Resources Research CenterUniversity of Hawaii, Honolulu

Whittier RB, Rotzoll K, Dhal S, El-Kadi AI, Ray C, Chen G, Chang D (2005a) Hawaii Source Water Assessment Program report, vol 2: Island of Hawaii. Prepared for Hawaii Dept. of Health, Water Resources Research Center, University of Hawaii, Honolulu

Whittier RB, Rotzoll K, Dhal S, El-Kadi AI, Ray C, Chen G, Chang D (2005b) Hawaii Source Water Assessment Program report, vol 6: Island of Molokai. Prepared for Hawaii Dept. of Health, Water Resources Research Center, University of Hawaii, Honolulu

Whittier RB, Rotzoll K, Dhal S, El-Kadi AI, Ray C, Chen G, Chang D (2006a) Hawaii Source Water Assessment Program report, vol 1: approaches used for the assessment. Prepared for Hawaii Dept. of Health, Water Resources Research Center, University of Hawaii, Honolulu

Whittier RB, Rotzoll K, Dhal S, El-Kadi AI, Ray C, Chen G, Chang D (2006b) Hawaii Source Water Assessment Program report, vol 5: Island of Maui. Prepared for Hawaii Dept. of Health, , Water Resources Research Center, University of Hawaii, Honolulu

 DOI 10.1007/s10040-009-0548-6